

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Ans: -

RAM Reseat – Step by Step Process

1 Prepare & Power Off  1. Shut down the laptop completely. 2. Disconnect the charger. 3. Place the laptop on a clean, flat surface.	2 Open the Back Panel  1. Remove the screws from the back panel. 2. Carefully lift off the panel using a plastic tool or your hands.	3 Locate the RAM Slots  1. Find the RAM slots on the motherboard. 2. In my laptop, there are 2 slots (one is occupied, the other is empty).	4 Remove the RAM Module  1. Gently push the side clips outward (left and right). 2. The RAM module will pop up. 3. Hold it by the edges and remove it.	5 RAM Module Removed  1. The module is now completely removed. 2. Keep it in a safe place.
6 Re-insert the RAM Module  1. Align the notch on the RAM with the slot. 2. Insert the module at an angle (~30°). 3. Press it down gently until the side clips lock it in place.	7 Check the Fit  1. Make sure the module is fully seated. 2. The clips should be locked and hold the RAM firmly.	8 Close the Laptop  1. Put the back panel in place. 2. Tighten all the screws.	9 Power On & Test  1. Connect the charger. 2. Turn on the laptop. 3. Check if the system boots properly and the RAM is recognized.	Issues Observed ✓ During the process: • No major issues faced. • RAM module was easy to remove and re-insert. • The clips were a bit tight, so I used gentle pressure. ✓ After re-inserting: • Laptop booted normally. • RAM was detected in BIOS (checked by pressing F2 at startup). • System is running smoothly without any error. ✓ Conclusion: • The RAM reseating process went smoothly. No issues were observed, and the laptop is working fine.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

Ans: -

GPU Troubleshooting Checklist

- **Identify the Symptom** – Check for no display, screen flickering, artifacts, crashes, or poor gaming performance.
- **Check Display Cable** – Make sure the HDMI/Display Port cable is firmly connected to the GPU and monitor.
- **Check GPU Power** – Ensure all required PCIe power connectors are properly connected.
- **Reseat the GPU** – Turn off the PC, remove the GPU, and firmly re-insert it into the PCIe slot.
- **Check GPU Driver** – Update or reinstall the correct graphics driver.
- **Check GPU Temperature** – Make sure the GPU is not overheating and its fans are working.
- **Test the GPU** – Run a game or graphics benchmark and check for crashes or display problems.
- **Confirm the Fix** – If the display works normally and games run without crashes or artifacts, the GPU issue is resolved.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Ans: -

1 Identify the Drive



SATA power and data cables connected to the SSD.

2 Unplug the Drive



Carefully unplug the SATA data cable (and power cable if needed).

3 Wait a Moment



Leave the drive disconnected for a few seconds.

4 Reconnect the Drive



Plug the SATA data cable back in (and power cable if needed).

5 Boot and Check BIOS



Enter BIOS and check if the SSD is detected again.

6 Check in Window



Verify the drive is visible and accessible in Windows

Before Reconnecting (Drive Disconnected)



During Reconnection (Error Message)



After Reconnecting (Drive Detected)



4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Ans: -

PC Randomly Restarts – Troubleshooting

- * **Component to check first: RAM**

Reason: Faulty or loose RAM can cause system instability, crashes, and random restarts.

Diagnostic Steps:

- * Shut down the PC and unplug the power.
- * Open the cabinet and locate the RAM.
- * Remove and **reseat the RAM** firmly.
- * Check the RAM for dust or physical damage.
- * If there are two RAM sticks, test **one stick at a time**.
- * Try different RAM slots if available.
- * Run **Windows Memory Diagnostic** to check for memory errors.
- * Restart and use the PC normally to confirm whether the random restarts have stopped.